02 · Final Report: writing guide

Owner suggestion: Member 1. **Length:** 6 to 10 pages. **Time:** 45 to 60 minutes if you reuse your Term 1 and 2 work.

Purpose

The brief calls it "a concise document that provides a brief background, problem statement, project objectives, scope and limitations, systems analysis (requirements) and testing and evaluation". It is the main document markers read. It tells the story: *what problem, what you promised, what you built, how you know it works, what is missing.*

What markers are likely to look for *(my inference from the brief)*

- A clear problem and measurable objectives, and whether the final system meets them.
- Honest scope and limitations (they know the brief asked for payments and microservices).
- Requirements that trace to features, and testing that traces to requirements.
- Evidence: diagrams, screenshots, test results.
- Project management thinking, not only technical description.

Structure (follow this order) and how to write each part

Cover page and contents

Title, course code PRM370/371/372S, group name and members, date. Add a short contents list.

1. Background (half a page)

What it is: the situation that made the project necessary. **Write about:** how students and the surrounding community currently buy and sell second-hand goods; why informal groups (WhatsApp, Facebook) fall short (no verification, scams, no records, not campus-specific); why a campus-and-community marketplace helps (affordability, sustainability, community). **Use:** the brief's introduction, your Term 1 charter, system facts in 01 section 1.

2. Problem statement (one paragraph)

What it is: one precise paragraph. Test it: who suffers, how, and what is the consequence? **Structure to follow:** the people affected, the problem, the impact, what is needed.

3. Project objectives (4 to 6 bullets)

What it is: what you set out to achieve, written so you can say yes or no at the end. Make them SMART (specific, measurable, achievable, relevant, time-bound). **Consider covering:** verified identity for

students; trustworthy ratings; easy listing and search; moderation and safety; mobile access (installable); community bulletin board; delivery on the project schedule. **Tip:** for each objective note later, in section 6, whether it was met and how you know.

4. Scope and limitations (1 to 1.5 pages)

- **In scope:** the features list from 01 section 3 (summarise in a table by module).
- **Out of scope / limitations:** be direct. Use the table in 01 section 8. For each: what the brief asked, what you did instead, why.
- **Constraints:** team of three, fixed timeline, student budget (free tools, no hosting or payment accounts), local deployment.
- **Assumptions:** students own their CPUT mailbox; users meet on campus and pay in person. **Why this matters:** a well-argued limitation protects your marks.

5. Systems analysis (2 to 3 pages)

5.1 Stakeholders and users (3 to 4 lines, refer to the stakeholder matrix).

5.2 Requirements. Explain functional requirements (what the system does) and non-functional requirements (how well: security, usability, performance, compatibility). Reuse your Term 2 backlog and user story IDs. Table to fill:

ID	Requirement	Type	Priority (MoSCoW)	Status (Done / Partly / Not done)	Evidence
FR-01	*Example:* Students register with a CPUT email	Functional	Must	Done	Test T1, T2
FR-02					

Requirement ideas drawn from the real system (adapt, merge, and match them to your backlog): CPUT-only student registration; vendor approval; login with expiring session; create/edit/delete listings with photos; search and filters; swap and Freecycle; chat per listing; mark sold to a named buyer; buyer confirms receipt; rating after confirmed receipt; wishlist with alerts; saved searches; in-app and push notifications; seller dashboard; report and block; admin moderation, suspension and appeals; bulletin board with Lost and Found; installable mobile app; passwords hashed; login rate limiting; images validated. **Not done from the brief:** cart and checkout, payment integration, escrow, AI fraud detection, 2FA, microservices, caching.

5.3 Design. Include two diagrams:

1. **Architecture diagram:** browser (Vue PWA) to REST API (Spring Boot) to database; mention the security filters and photo storage. Draw it from 01 section 4.
2. **Data model (ER diagram):** the 13 tables and relationships in 01 section 5. Optional: a sequence diagram of the verified sale (seller marks sold, buyer confirms, buyer rates) and 2 or 3 screenshots from `demo/screenshots/`.

5.4 Technology choices. One short table: technology, why chosen (for example: Vue for fast mobile-first UI and PWA; Spring Boot for secure, structured APIs; MySQL for relational data like users-listings-

ratings).

6. Testing and evaluation (2 pages)

6.1 Approach. Describe your testing: automated unit tests, manual scenario testing, live checks. Say who tested.

6.2 Results. Facts you can report:

- 25 automated backend tests, all passing. They cover the CPUT-email rule (including lookalike domains), vendor approval, password hashing, duplicate username, wrong password, token validity/expiry/forgery, image validation, login rate limiting and the production secrets check.
- Live checks: brute-force login blocked on the 6th attempt; oversized and fake image uploads rejected; a 3.8 MB photo accepted.
- **A defect found and fixed:** six simultaneous "mark sold" requests reduced stock by one instead of six (a race condition). Fixed with a database lock, retested (50 to 44).
- Not automated: marketplace, chat and admin flows were checked manually.

6.3 Test cases. Run these yourselves and fill in "Actual" and "Pass/Fail" honestly:

ID	Scenario	Steps	Expected	Actual	Pass?
T1	Student must use CPUT email	Register STUDENT with a gmail address	Rejected with message		
T2	Student registers with CPUT email	Register with x@mycput.ac.za	Account created		
T3	Vendor needs approval	Log in as quickfix_it, try to create a listing	Blocked until admin approves		
T4	Wrong-password lockout	6 wrong logins for one username	6th returns "Too many attempts"		
T5	Create listing with photo	Dashboard, New Listing, add a 3 MB JPEG	Listing shows photo		
T6	Oversized photo	Add a 5 MB JPEG	Clear error, not saved		
T7	Chat per listing	Buyer messages seller, seller replies	Both see messages		
T8	Sale to a named buyer	Seller marks sold to buyer	Listing shows Sold, buyer notified		
T9	Rating needs confirmed receipt	Buyer tries to rate before confirming	Blocked		
T10	Rating after confirmation	Buyer confirms then rates 5 stars	Rating saved, seller average updates		
T11	Cannot rate twice	Rate the same sale again	Rejected		
T12	Report and moderate	Report a listing; admin reviews	Appears in queue, admin can act		
T13	Suspension works at once	Admin suspends a logged-in user	User blocked on next request		
T14	Wishlist alerts	Heart an item; seller drops price	Notification arrives		
T15	Install as PWA	Chrome install prompt	Installs and opens		

6.4 Evaluation against objectives. A table: objective, met (yes / partly / no), evidence. Then 2 or 3 paragraphs: what worked, what did not, what you would improve.

7. Conclusion and recommendations (half a page)

Summarise what was delivered, how it answers the problem, the key limitations, and next steps (email verification, payment gateway with hosted checkout, 2FA for high-value purchases, cloud photo storage, privacy policy and account deletion, independent penetration test, more automated tests).

References

List anything you quote or use (the course textbook, Spring Boot and Vue documentation, photo credits). Use the referencing style your department requires.

Common mistakes to avoid

- Claiming features the system does not have (payments, escrow, AI, microservices).
- Describing code instead of explaining the *project* (this is a project management course).
- Objectives that cannot be measured.
- Test results you did not actually run.
- Copying text from the brief, the README or another document without rewriting.

Checklist

- [ ] Background, problem, objectives, scope, limitations, analysis, testing, conclusion all present
- [ ] Two diagrams (architecture, data model) and 2 to 3 screenshots
- [ ] Requirements table with status and evidence
- [ ] Test table filled with real results
- [ ] Limitations stated honestly with reasons
- [ ] Names, features and numbers match the other documents